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Plug-flow reactor and shock-tube study of the oxidation of very fuel-rich natural gas/DME/O_2 mixtures

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ABSTRACT

A polygeneration process with the ability to provide work, heat, and useful chemicals according to the specific demand is a promising alternative to traditional energy conversion systems. By implementing such a process in an internal combustion engine, products like synthesis gas or unsaturated hydrocarbons and very high exergetic efficiencies can be obtained through partial oxidation of natural gas, in addition to the already high flexibility with respect to the required type of energy. To enable compression ignition with natural gas as input, additives such as dimethyl ether are needed to increase the reactivity at low temperatures. In this study, the reaction of fuel-rich natural gas/dimethyl ether (DME) mixtures is investigated to support the further development of reaction mechanisms for these little studied reaction conditions. Temperature-resolved species concentration profiles are obtained by mass spectrometry in a plug-flow reactor at equivalence ratios $\phi = 2$, 10, and 20, at temperatures between 473 and 973 K and at a pressure of 6 bar. Ignition delay times and product-gas analyses are obtained from shock-tube experiments, for $\phi = 2$ and 10, at 710 – 1639 K and 30 bar. The experimental results are compared to kinetic simulations using two literature reaction mechanisms. Good agreement is found for most species. Reaction pathways are analyzed to investigate the interaction of alkanes and DME. It is found that DME forms radicals at comparatively low temperatures and initiates the conversion of the alkanes. Additionally, according to the reaction pathways, the interaction of the alkanes and DME promotes the formation of useful products such as synthesis gas, unsaturated hydrocarbons and oxygenated species.

1. Introduction

The increasing use of fluctuating renewable energy requires the development of flexible energy conversion systems. One possible concept is polygeneration that can provide different types of energy depending on demand. Atakan [1] and Hegner et al. [2] performed modeling studies of a fuel-rich operated gas turbine and a homogenous-charge compression-ignition (HCCI) engine and found that the simultaneous – but variable – provision of work and chemicals as well as heat in a thermochemical process can lead to very high exergetic efficiencies. They compared the polygeneration process with combined individual processes to provide work, heat, and hydrogen and highlighted the higher efficiency of the polygeneration process. In both studies, methane (CH_4) was used as fuel. Gossler and Deutschmann [3] showed that H_2 yields of more than 90 % can be achieved theoretically using fuel-rich HCCI conditions with a fuel mixture of 99 % CH_4 and 1 % propane (C_3H_8). More recently, such processes were investigated experimentally in a rapid-compression machine (RCM) [4] and an HCCI engine [5,6]. In these studies, it was shown that, besides providing a considerable amount of work, high yields of synthesis gas are produced under slightly fuel-rich conditions ($\phi \approx 2$). The product mix of work and chemicals resulted in high exergetic efficiencies of the investigated processes.

The above-mentioned conversion of methane towards valuable chemicals preferentially takes place at high equivalence ratios (typically $\phi >> 2$), moderate temperatures, and high pressures. At these conditions, the fuel (e.g. CH_4 or natural gas) is partially oxidized such that synthesis gas, oxygenated hydrocarbons, or larger hydrocarbons compared to the reactants are formed. As HCCI engines are based on autoignition and CH_4 is relatively inert, more

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Table 1
Recent experimental studies on the ignition delay times of natural gas and DME mixtures with their corresponding experimental conditions (ST: shock tube, RCM: rapid-compression machine, CVB: constant volume bomb).

Mixtures	p / bar	T / K	ϕ	Facility	Ref.
CH_4/C_2-C_5	0.2	1400-2000	1	ST	[43]
CH_4	3-300	1200-2100	0.2-2	ST	[44]
CH_4/C_2-C_4	4	1200-1850	0.2-0.4	ST	[45]
CH_4/C_2-C_4	3	1400-2000	0.2	ST	[46]
CH_4/C_3	2.5	1300-1600	1	ST	[47]
CH_4/C_2-C_4	3-15	1300-2000	0.45-1.25	ST	[48]
CH_4	9-480	1410-2040	0.5-4	ST	[49]
DME	13-40	650-1300	1	ST	[23]
DME	3.5	1200-1600	0.5-2	ST	[22]
CH_4	40-260	1040-1500	0.4-6	ST	[50]
CH_4	35-260	1040-1600	0.4-6	ST	[51]
DME	0.83-2.9	900-1900	Pyrolysis	ST	[52]
CH_4/C_2-C_3	5-20	900-1700	0.5-1	ST	[53]
CH_4/C_2-C_3	3-13	1485-1900	0.5-2	ST	[54]
CH_4	3-450	1200-1700	0.5	ST	[55]
CH_4	16-40	1000-1350	0.7-1.3	ST	[56]
CH_4/C_2-C_3	16-40	900-1400	1	ST	[57]
CH_4/C_2-C_3, H_2	0.54-30	1091-2001	0.5	ST	[58]
CH_4/C_2-C_5	15.5-27.5	786-1107	0.5	ST	[59]
CH_4/C_3H_8	10-30	740-1550	0.3-3	ST, RCM	[60]
$CH_4/C_2H_6/C_3H_8$	1-50	770-1580	0.5-2	ST, RCM	[61]
$CH_4/C_2/C_3$	13-21	850-925	0.6-1	RCM	[62]
CH_4/H_2	15-70	950-1060	0.5-1	RCM	[63]
DME	10-20	615-735	0.43-1.5	RCM	[64]
$CH_4/C_2H_6/H_2$	1-16	900-1800	0.5-1	ST	[65]
DME	1.6-6.6	1175-1900	0.5-3	ST	[66]
CH_4/DME	1-10	1134-2105	1	ST	[37]
CH_4/C_2H_6	1-31	1154-2250	0.5-2	ST	[67]
DME/C_3H_8	20	1100-1500	0.5-2	ST	[38]
C_2H_6/DME	2-20	1100-1500	0.5-2	ST	[41]
CH_4	1-10	1350-1950	0.5-2	ST, CVB	[68]
CH_4/DME	7-41	600-1600	0.3-2	ST, RCM	[28]
CH_4/DME	1-1.6	1650-2050	2	ST	[14]
DME	0.55-1.2	1075-1860	0.5-1	ST	[69]

reactive species like *n*-heptane [5] or dimethyl ether (DME) [4,7] can be added to the fuel to reach autoignition conditions at the end of the compression stroke. Validated reaction mechanisms for such mixtures that could support the development of reaction concepts under these uncommon conditions are only slowly becoming available and require further improvement. The development requires experiments under well-controlled boundary conditions.

For CH_4, the homogenous partial oxidation has been investigated in flow reactors [8–16] or jet-stirred reactors [17]. The investigated conditions cover pressures from 1 – 100 bar, temperatures between 600 and 1800 K and equivalence ratios ranging from 0.06 to 100. Most of these studies focus on the development of reaction mechanisms for fuel-rich combustion, whereas the work of Rytz and Baiker [8] specifically focused on the optimization of the generation of methanol. They found that the selectivity towards methanol increases with increasing pressure and decreases with increasing temperature or increasing oxygen concentration.

Ignition delay times of homogenous fuel/air mixtures were measured for fuel-rich methane and natural gas in shock tubes or rapid-compression machines, as summarized in Table 1; species concentration profiles were measured less frequently. El Bakali et al. [18], Tan et al. [19], and Kaczmarek et al. [20] measured species concentration profiles as a function of temperature for various natural gas mixtures in jet-stirred and flow reactors. The experiments cover a range of ϕ = 0.5 – 20, p = 1 – 10 bar and T = 473 – 1240 K. In addition, El Bakali et al. [18] also investigated the oxidation of natural gas in a stoichiometric premixed flame at low pressure (10.6 kPa).

The kinetics of DME have been investigated extensively in the last two decades. In addition to ignition delay time measurements in different experimental configurations (Table 1), several reactor studies have been performed. Dagaut et al. [21,22] investigated the oxidation of DME in a jet-stirred reactor at ϕ = 0.25 – 2, p = 1 – 10 bar and T = 550 – 1300 K and proposed one of the first reaction mechanisms, using also shock-tube ignition delay times (IDTs) from Pfahl et al. [23]. Curran et al. [24] improved this reaction mechanism by adjusting the rate constants for unimolecular fuel decomposition and methoxy–methyl radical β-scission and adding a new low-temperature oxidation scheme. Further modifications on the reaction mechanism were done by Fischer et al. [25] and Curran et al. [26]. They used additional data, obtained in jet-stirred and flow reactor experiments under oxidizing (ϕ = 0.2 – 4.2, p = 1 – 18 bar, T = 800 – 1300 K) and pyrolyzing (2.5 bar, 1060 K) conditions as well as IDTs from shock-tube experiments [23]. They observed formic acid as a major intermediate of DME oxidation and added reaction pathways leading to formic acid for better predictions of the experimental data [26].

Zhao et al. [27] proposed a model after revising the rate constant of the unimolecular decomposition reaction of DME and considering flow reactor experiments under pyrolyzing conditions. More recently, Burke et al. [28] incorporated measured rate constants and high-level calculations for the reactions of DME and applied a pressure-dependent treatment to the low-temperature chemistry of DME within their detailed kinetics mechanism. They also used new IDTs with respect to CH_4, DME and their mixtures [28] for validation. More reactor experiments with respect to the oxidation of DME can be found in [29–35]. Except the studies of Hashemi et al. [34] and Porras et al. [35], all of these studies have been performed at atmospheric pressure. Hashemi et al. [34] extended the available database regarding the oxidation of DME to pressures up to 100 bar and equivalence ratios up to 20. Additionally, they performed pyrolysis experiments and experiments with CH_4/DME blends. They also proposed a kinetics model for DME/CH_4 oxidation and concluded that the addition of DME to the CH_4 fuel has a promoting effect on the reaction onset temperature of CH_4 and that the effect is more pronounced in extremely fuel-rich conditions (ϕ = 20). Similar results can be found in Refs. [10,14], where the addition of DME was also found to be more effective than the addition of C_2H_6. Porras et al. [35] extended the experimental database with respect to fuel-rich oxidation (ϕ = 2 – 10) of CH_4/DME mixtures to the intermediate pressure range (6 – 30 bar) with flow-reactor, rapid-compression machine, and shock-tube experiments with special emphasis on polygeneration processes. The authors also proposed a new kinetics model for the description of the reaction of these mixtures

DME can accelerate ignition [36–39] if it is added to methane or propane and it increases the flame speed of methane/air mixtures [36,40]. But, with respect to ethane, it was shown that the DME blending ratio shows a nonlinear effect on the IDTs due to interactions between DME and the radical pool provided by ethane oxidation [41]. More fundamental experiments regarding the oxidation of natural gas/DME mixtures showing such interactions are rare. The present work is intended to fill this gap by extending the study of these mixtures to high equivalence ratios.

This study presents ignition delay times and species profiles as a function of temperature for different natural gas (90 % CH_4, 9 % C_2H_6, 1 % C_3H_8)/DME blends at equivalence ratios of 2, 10, and 20 by combining complementary results of shock-tube (ϕ = 2 and 10, p = 30 bar, T = 710 – 1630 K) and flow-reactor measurements (ϕ = 2, 10, and 20, p = 6 bar, T = 473 – 973 K). The results are compared to predictions of two reaction mechanisms from literature [35,42] and a modified mechanism to assess their ability of describing the interaction between DME and natural gas under uncommon reaction conditions.

The experimental, modeling, and results part of the paper is divided into three chapters. In section 2, the experimental methods

Table 2
Reaction conditions (PFR: plug flow reactor, ST: shock tube).

CH_4 Mol%	C_2H_6 Mol%	C_3H_8 Mol%	DME Mol%	O_2 Mol%	Ar Mol%	N_2 Mol%	Ne Mol%	p bar	T K	ϕ -	Exp. -
4.065	0.406	0.045	0.238	5.246	90	–	–	6	473-973	2	PFR
3.813	0.381	0.042	0.471	5.292	90	–	–	6	473-973	2	PFR
7.004	0.700	0.078	0.410	1.808	90	–	–	6	473-973	10	PFR
6.613	0.661	0.073	0.816	1.836	90	–	–	6	473-973	10	PFR
7.700	0.770	0.086	0.450	0.994	90	–	–	6	473-973	20	PFR
7.281	0.728	0.081	0.899	1.011	90	–	–	6	473-973	20	PFR
14.330	1.433	0.159	–	17.236	–	66.842	–	30	950-1630	2	ST
13.396	1.340	0.149	0.783	17.288	–	67.044	–	30	850-1450	2	ST
12.492	1.249	0.139	1.542	17.338	–	67.240	–	30	710-1480	2	ST
44.245	4.425	0.492	–	10.422	–	38.416	2.000	30	930-1540	10	ST
41.176	4.118	0.458	2.408	10.627	–	39.213	2.000	30	860-1500	10	ST

are presented and relevant details of the individual experimental arrangements are given. Section 3 deals with modeling. In section 3.1, sensitivity analyses are shown that form the basis for the modification of the mechanism of Porras et al. [35]. The most sensitive reactions at different conditions are evaluated and the modifications to the mechanism are described. Sections 3.2 and 3.3 briefly outline the simulation methods used.

Section 4 discusses the results, starting with the shock-tube measurements. While section 4.1.1 presents ignition delay times for CH_4/air, natural gas/air, and natural gas/DME/air mixtures at $\phi = 2$ and 10 and 30 bar, section 4.1.2 shows the products formed after ignition for natural gas/air and natural gas/DME/air mixtures at $\phi = 10$ and 30 bar. In section 4.2, the results of the reactor experiments are presented with a focus on the influence of DME on the fuel conversion and product composition. In chapter 4.2.1, figures presenting all experimental results compared to the predictions of the different mechanisms are shown. These results are further discussed in the following sub-sections. The first section examines the fuel conversion and the formation of three selected types of product species: synthesis gas (H_2, CO), unsaturated hydrocarbons (C_2H_4, C_3H_6), and oxygenated species (CH_3OH, CH_2O). The reaction pathways towards the individual products are analyzed in detail and possible reasons for deviations of the model predictions are discussed. The individual sections are briefly summarized at the end of each chapter to highlight the core statements. In the last result chapter, the maximum yields of the main species are compared for all mixtures in a single diagram and the overall findings are discussed without further analysis. Together with the species profiles at the beginning of the chapter, readers who are primarily interested in the product composition can view the main results without having to go into further details.

2. Experimental methods

The investigated reaction conditions are shown in Table 2. Details of both experimental arrangements are given below.

2.1. Shock tube

A high-pressure shock tube for post-ignition peak pressures up to 500 bar with a constant inner diameter of 90 mm and lengths of the driver and driven sections of 6.4 and 6.1 m, respectively, was used for ignition delay time (IDT) and product composition measurements. The maximum test time is extended up to 15 ms by driver-gas tailoring with helium as the main driver gas component and Ar addition to match the acoustic impedance of the driver gas with the one of the test gas. The driver gas was mixed *in situ* by using two high-pressure mass flow controllers (Bronkhorst Hi-Tec flowmeter F-136AI-FZD-55-V and F-123MI-FZD-55-V). The driver gas composition depends on the test gas mixture and the Mach number and was calculated prior to the experiment using formulas by Oertel [70] and Palmer and Knox [71]. Test gas mixtures were prepared manometrically in a mixing vessel and stirred for at least one hour to ensure homogeneity. The experiment is described in more detail in Ref. [72].

The temperature T_5 and pressure p_5 behind the reflected shock waves were computed from the incident shock velocity using a one-dimensional shock-wave model. The shock-wave velocity was measured over three intervals using four piezoelectric pressure transducers (PCB 112-B05) and the related temperature uncertainty of ±15 K was determined by the method of Petersen et al. [73] that considers the uncertainty of the position of the pressure transducer, the time when the incident shock wave passes the pressure transducer, and the error caused by the interpolation of the incident shock-wave speed to that at the end flange. The determined error represents an average value for the temperature range of our measurements. Band-pass-filtered (308±5, 431±5 nm) emission from OH* and CH* chemiluminescence was monitored through windows in the sidewall 15 mm from the end flange with fiber-coupled Hamamatsu 1P28 photomultipliers. The pressure was recorded at the same position using a piezoelectric pressure transducer (PCB-112B05) that was shielded with a thin high-temperature silicone layer. Ignition delay times were defined as the interval between the rise in pressure due to the arrival of the reflected shock wave at the measurement port and the extrapolation of the steepest increase in emission to its zero level on the time axis. The uncertainty of the determination of ignition delay times from the chemiluminescence profiles is about ±10 μs for IDTs < 100 μs, ±30 μs for IDTs of about 1000 μs and ±100 μs for IDTs around 10 ms. These values were determined considering the uncertainties of the extrapolation of the steepest increase of the measured chemiluminescence intensity and the uncertainties of the blast-wave correction [51].

Gas samples were taken from the shock tube with a fast-acting valve (Parker Hannifin Pulse Valve Series 9 with IOTA ONE® driver) that was integrated in the end flange and connected to a 50-ml sample vessel. The valve was opened 14 ms after the arrival of the reflected shock wave for 10 ms. The samples were analyzed with GC/MS (Agilent Technologies GC System 7890 A, MSD 5975C) using an Agilent J&W HP-PLOT Q column. Each mixture used in the experiments at $\phi = 10$ contained 2 % Ne as internal standard. All mole fractions were determined relative to this internal standard so that the measured species mole fractions are independent on the inlet pressure in the sample and the accuracy is limited by the errors in calibration, the preparation of the fuel/air mixture, and by the uncertainty of the GC measurement. Calibration mixtures with various mole fraction levels were prepared in a 50-L mixing vessel using very precise pressure transducers. We estimate that the error of the calibration factor caused by the mixture preparation and the GC measurement uncertainty is below 5%. For all substances, linear behavior with no offset between signal and mole fraction was observed. Another 5% error is related to uncertainties in the Ne mole fraction in the fuel/air mixture. The low uncertainty ($<$ 10% in total) was confirmed by measurements of the fuel/air mixtures at temperatures where no oxidation occurs and by calculating the carbon balance, which was found to be 1±0.05 at low temperatures, where oxidation but no formation of larger PAHs occurred.

2.2. Plug-flow reactor

In this study, a flow reactor coupled to a time-of-flight mass spectrometer (see supplementary material or [20]) is used to investigate the partial oxidation of natural gas/DME mixtures at a pressure of 6 bar and temperatures between 473 and 973 K. A constant volumetric flow rate of 280 sccm (standard conditions: $T = 273.15$ K, $p = 1$ bar) was used to approximate plug-flow conditions according to the model of Levenspiel [74]. The residence time varied between 14.5 s at 473 K and 7.6 s at 973 K, considering the complete reactor length. Details of the experimental setup are given elsewhere [14,75] and only a brief description is provided here.

The reactor consists of a 65 cm long quartz tube, surrounded by a stainless-steel tube for safety. A 40-cm long isothermal reaction zone (±10 K) is established by temperature-controlled heating tapes, wrapped around a copper shell that encloses the stainless-steel tube. The gap between the stainless-steel and the quartz tube is sealed at the in- and outlet of the reactor to prevent gases from entering this gap and reacting with the steel wall. The reactants are metered by calibrated mass flow controllers and premixed at room temperature before entering the reactor, while the pressure is regulated manually by a heated needle valve downstream of the reactor. Pressure fluctuations, measured by a pressure transducer upstream of the reactor, are less than 20 mbar. A high argon dilution is used to minimize the temperature rise due to exothermal reactions.

After passing the reaction zone, the product gases are guided through the needle valve into a time-of-flight mass spectrometer (TOF-MS, Kaesdorf) via molecular-beam extraction for product analysis. The products are ionized by electron ionization at 17 eV to minimize fragmentation. Because fragmentation could not be avoided completely, it was considered for all directly calibrated species in data analysis. The mass resolution of the spectrometer is $m/\Delta m \approx 2000$. The TOF-MS enabled the detection of stable species, formed in the partial oxidation process. The mole fractions of the reactants (CH_4, C_2H_6, C_3H_8, DME, O_2) as well as those of CO, CO_2, C_2H_4, C_3H_6, CH_2O and CH_3OH were measured. The uncertainties with respect to mole fractions of gaseous and liquid species are estimated to be around 10 and 20 %, respectively.

3. Modeling

Reaction mechanisms of Porras et al. [35] and Zhou et al. [42] were used for the simulation of the shock-tube and flow-reactor experiments. The reaction mechanism of Porras et al., called PolyMech (in the following PolyMech1.0), was developed for the prediction of very fuel-rich CH_4 and CH_4/additive mixtures, by combining literature mechanisms developed for the pyrolysis and combustion of CH_4, as well as an ethanol and a DME submechanism (by Zhao et al. [27]). Also, several rate constants were adjusted within their documented uncertainty range for a better prediction of ignition delay times and species profiles as a function of temperature, obtained by shock-tube, RCM and flow-reactor experiments at high equivalence ratios. It covers the oxidation of H_2, C_2H_2, C_1–C_3 alkanes, and alkenes as well as oxygenated C_1–C_2 species.

The reaction mechanism of Zhou et al. [42], also known as AramcoMech3.0, was developed at NUI Galway and covers C_1–C_4-chemistry, including oxygenated species and peroxide chemistry. The included DME mechanism is that of Burke et al. [28]. Similar to the mechanism of Porras et al. [35], the AramcoMech3.0 [42] shows a hierarchical structure, ranging from an H_2, to butene and butadiene mechanisms. It was validated by a large experimental dataset, comprising data from shock tubes, RCMs, flames, and various types of reactors making it well-suited for the simulations of the mixtures investigated in this study.

3.1. Model improvement

To improve the PolyMech predictions of experimental data, especially with species profiles of CO, CO_2, H_2, C_2H_2, C_2H_4, C_3H_8, and C_6H_6, some rate constant expressions were updated by more recent ones and missing pathways were added to the mechanism. The reactions were identified by integrated sensitivity analyses and comparative reaction path analyses using the PolyMech1.0 and AramcoMech3.0. Here, only the analyses performed with the PolyMech1.0 are shown. Sensitivity analyses performed with the updated PolyMech (PolyMech2.0) and the AramcoMech3.0 can be found in the supplementary material. The AramcoMech3.0 was used for comparison because its predictions showed a good agreement with the experimental data, it contains current rate constants, and is one of the most detailed mechanisms with respect to C_1–C_4-chemistry. This is also the reason why the mechanism was not modified in contrast to the PolyMech1.0, which is an in-house mechanism with a very limited number of species and reactions and specifically made for the polygeneration conditions investigated here. Modified or added reactions are listed in Table 3. Sensitivity coefficients (S_i) are defined as

$$S_{ij} = \frac{\partial x_i}{\partial k_j}\frac{k_j}{x_i} \tag{1}$$

with x_i and k_i representing the mole fraction and rate constant of the specific species and reaction, respectively.

Figure 1 shows the results of the sensitivity analysis for CO_2. It is found that the reaction of CO and hydroperoxyl radicals (HO_2) is the most sensitive one with an increasing sensitivity at higher equivalence ratios and a decreasing sensitivity at higher temperatures. Its rate constant expression was replaced by the expression implemented in the AramcoMech3.0 and proposed by You et al. [76], showing much lower rates at temperatures higher than 800 K. The reaction of methyl (CH_3) and HO_2 radicals, showing the second highest sensitivity under the investigated conditions, was updated to the values given by Jasper et al. [77]. These modifications resulted in a very good prediction of the measured CO and CO_2 mole fractions in the plug-flow reactor. To further improve the predictions for the shock-tube conditions, the rate expression for the reaction of CO and hydroxyl radicals (OH) was adapted from Joshi and Wang [78]. Only the low-pressure-limit rate coefficient, which was fitted by a sum of two modified Arrhenius equations (reactions 2 and 3), was implemented in PolyMech1.0, as the authors stated that it is applicable for most combustion applications. The rate constants for reactions of HCCO and O_2 (reactions 5 and 6) were updated to the values proposed by Klippenstein et al. [79].

With respect to CH_3OH, the sensitivity analysis was performed for 623 and 773 K, two different DME amounts and $\phi = 2$ (Fig. 2), at which the largest deviation between experiment and simulation was found. The modeling predictions are most sensitive to reactions involving DME at 623 K, because of its low-temperature chemistry. The sensitivities are much higher for the 5 % DME mixture compared to the 10 % DME mixture. Because of a very good prediction of DME mole fractions at all conditions, the DME chemistry set was not modified. The CH_3OH mole fraction is also

Table 3
Modified and added reactions of the PolyMech2.0. The rate constants are in the form of k = AT^{n}exp(-E/(RT)). Units are mol, cm, K, s, and kJ.

Reaction		A	n	E_A	Ref.
1	$CO + HO_2 = CO_2 + OH$	1.57×10^5	2.18	75.061	[76]
2	$CH_3 + HO_2 = CH_3O + OH$	1×10^{12}	0.269	-2.8765	[77]
3	$CO + OH = CO_2 + H$	7.046×10^4	2.053	-1.489	[78]
4	$CO + OH = CO_2 + H$	5.757×10^{12}	-0.664	1.389	[78]
5	$HCCO + O_2 = CO + CO_2 + H$	4.78×10^{12}	-0.142	4.812	[79]
6	$HCCO + O_2 = CO + CO + OH$	1.91×10^{11}	-0.020	4.268	[79]
7	$CH_3O + CH_2O = CHO + CH_3OH$	6.62×10^{11}	0	9.5981	[80]
8	$CH_3OH + HO_2 = CH_3O + H_2O_2$	1.22×10^{12}	0	83.976	[81]
9	$CH_3OH + O_2 = CH_3O + HO_2$	3.58×10^4	2.27	178.927	[82]
10	$CH_2O + OH = CHO + H_2O$	7.82×10^7	1.63	-4.414	[83]
11	$CH_2O + CH_3 = CHO + CH_4$	3.83×10^1	3.36	18.041	[84]
12	$CH_2O + H = CHO + H_2$	5.860×10^3	3.13	6.3353	[85]
13	$C_2H_4(+M) = C_2H_2 + H_2(+M)$	8.0×10^{12}	0.44	371.414	[86][a]
14	$C_2H_4 + OH = C_2H_3 + H_2O$	1.30×10^{-1}	4.2	-3.598	[87]
15	$C_2H_5 + HO_2 = CH_3CH_2O + OH$	3.00×10^{13}	0	0	[88]
16	$C_2H_4 + H = C_2H_3 + H_2$	1.266×10^5	2.752	48.739	[89]
17	$CH_3 + CH_3\ (+M) = C_2H_6\ (+M)$	9.455×10^{14}	-0.538	0.5654	[90,91][a]
18	$C_3H_8 + OH = N\text{-}C_3H_7 + H_2O$	2.732×10^7	1.811	3.633	[92]
19	$C_3H_8 + OH = I\text{-}C_3H_7 + H_2O$	1.975×10^7	1.751	0.266	[92]
20	$PC_3H_4 + H = C_2H_2 + CH_3$	2.830×10^8	1.74	32.202	[93]
21	$CH_4\ (+M) = CH_3 + H + (M)$	2.105×10^{16}	0	439.00	[94][a]

[a] See supplementary material for low-pressure limits and Troe parameters.

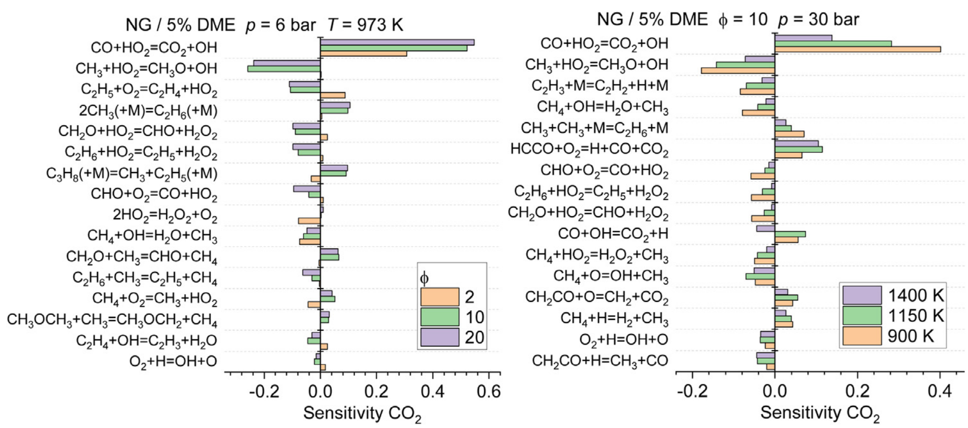

Fig. 1. Sensitivity coefficients of the CO_2 mole fraction for selected plug-flow (left) and shock-tube (right) conditions using the PolyMech1.0 [35]. Coefficients were integrated up to the sampling event of each experiment.

sensitive to H-abstraction from CH_4 and C_2H_6 by OH radicals, especially at low temperature. The rate constant expression for CH_4+OH, implemented in the PolyMech1.0, was adapted from Baulch et al. [84] with a reliability given by $\Delta\log k = \pm0.1$ over the range 250–350 K, rising to ±0.2 at 800 K and ±0.3 at 2400 K. Taking these error limits into account, the expression is in very good agreement to the more recent one proposed by Srinivasan et al. [95] and was congruously not modified. The same applies to the expression for C_2H_6+OH that only shows a maximum deviation of 9.78 % at 570 K, compared to the expression proposed by Krasnoperov et al. [96]. This rate expression was obtained by new shock-tube studies and shows an excellent agreement to literature data. The rate expression for the reaction of methoxy (CH_3O) radicals and formaldehyde (CH_2O), however, was changed to the one implemented in the AramcoMech3.0 and originally proposed by Fittschen et al. [80] because it improved the model predictions by 1.5 to 3 times higher rates at the relevant temperatures. As the results obtained by the AramcoMech3.0 still showed a better agreement to the experiments, a reaction pathway analysis was performed to identify important pathways yielding CH_3OH. The two most important reactions were added to the PolyMech as reactions 8 and 9 (see Table 3).

For CH_2O, the PolyMech1.0 slightly underpredicts the mole fractions at temperatures between 600 and 800 K at all equivalence ratios. In Fig. 2, the sensitivity analysis performed at 773 K for all 5 % DME mixtures, identifies only three reactions with high sensitivities at all ϕ. These reactions are the already discussed H-abstraction from CH_4 by OH and H-abstractions from CH_2O by OH (reaction 10) and CH_3 (reaction 11). The rate expression of reaction 10, originally adapted from Warnatz [97], was replaced by the more recent rate expression proposed by Vasudevan et al. [83]. They combined new high-temperature shock-tube measurements with low-temperature measurements by Sivakumaran et al. [98]. The three-parameter fit covers the temperature range between 200 and 1670 K and shows good agreement to experimental data at intermediate temperatures [83]. Comparing both rate expressions, more than two times lower rates are found for the new expression throughout the investigated temperature range, yielding higher CH_2O mole fractions. Regarding the rate expression of reaction 11, only the *A* factor was reduced by 20 % and thus changed to its original value, recommended by Baulch et al. [84]. Lastly, reaction 12 was updated to the values proposed by Wang et al. [85]. The same three rate expressions are also implemented in the AramcoMech3.0.

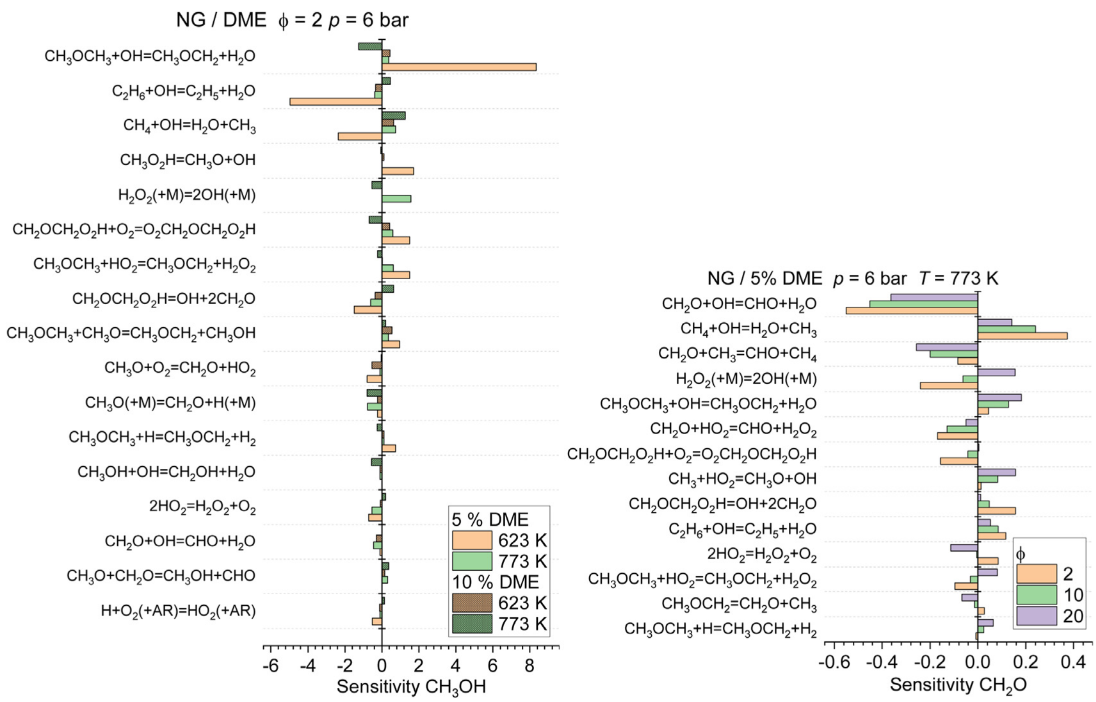

Fig. 2. Sensitivity coefficients of the CH_3OH (left) and CH_2O (right) mole fractions for selected plug-flow conditions using the PolyMech1.0 [35]. Coefficients were integrated up to the sampling event of each experiment.

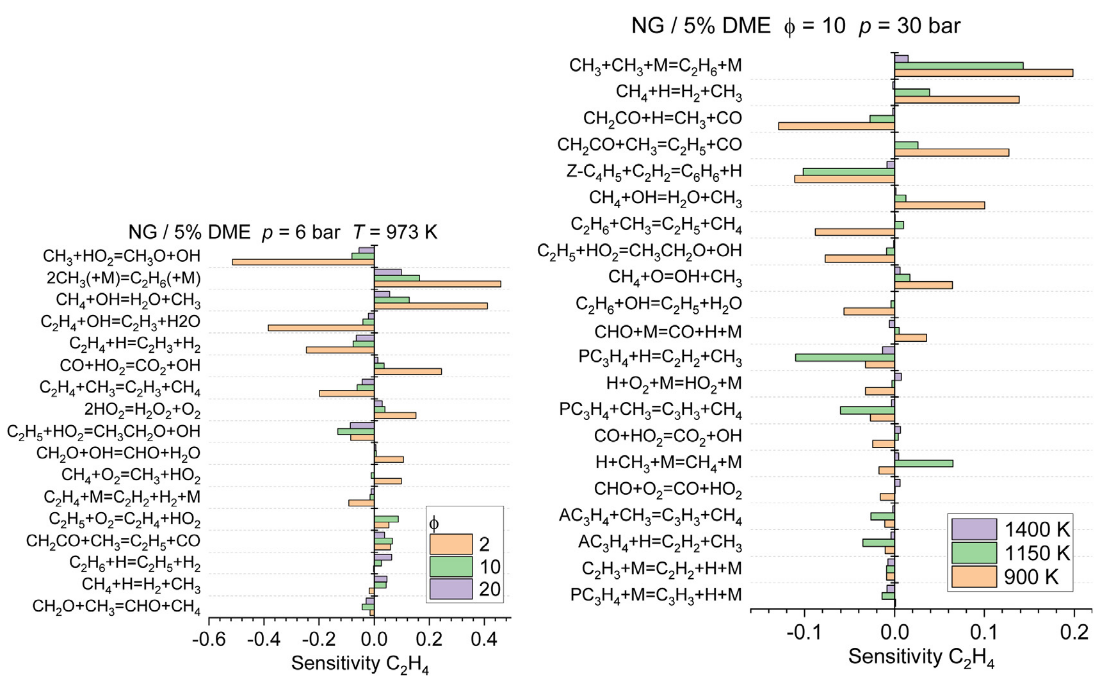

Fig. 3. Sensitivity coefficients of the C_2H_4 mole fraction for selected plug-flow (left) and shock-tube (right) conditions using the PolyMech1.0 [35]. Coefficients were integrated up to the sampling event of each experiment.

The sensitivity analysis for C_2H_4 (Fig. 3) reveals that, besides reaction 2 (only plug flow), the recombination of CH_3 radicals (reaction 17) is the most sensitive reaction for the selected plug-flow and shock-tube conditions. The rates of the recombination reaction were replaced by the values proposed by Wang et al. [91] and Klippenstein et al. [90], using the high-pressure limit of Klippenstein et al. and the low-pressure limit as well as the Troe parameters of Wang et al. Implementing the high-pressure limit of Klippenstein et al. [90] resulted in 1.2 times higher rates at 700 K rising to 1.4 times higher rates at 1400 K in the investigated pressure range compared to the rates obtained by the expression of Wang et al. [91], which is implemented in the AramcoMech3.0. In addition, the reactions of C_2H_4 with OH (reaction 14) and H (reaction 16) were updated to the values proposed by Senosiain et al. [87] and Huynh et al. [89], especially to improve the model predictions at the investigated plug-flow conditions. The new rate constant expressions lead to slightly lower and up to five times lower rates, respectively, compared to the rate expressions integrated in the PolyMech1.0. To slightly improve model predictions at lower temperatures, the rate expression for the reaction of C_2H_5 and HO_2 was taken from Bozzelli et al. [88].

With respect to the shock-tube analysis, reactions of acetylene (C_2H_2) and propargyl radicals (C_3H_3) are also shown to be very sensitive to the prediction of C_2H_4. Both species are precursors of benzene (C_6H_6), which is found in the shock-tube experiments in considerable amounts. As the PolyMech1.0 contains only three reactions describing the formation of C_6H_6 and none for its consumption, much higher mole fractions of C_6H_6 are predicted compared to the experiments. To improve this, the complete C_6H_6 sub-mechanism of the reaction mechanism of Cai and Pitsch [99] including the formation of some higher polycyclic aromatic hydrocarbons (PAHs) was adapted and added to the PolyMech. The benzene/PAH sub-mechanism of Cai and Pitsch predicts C_6H_6 mole fractions with very good agreement for experimental shock-tube data for CH_4, CH_4/diethyl ether, CH_4/*n*-heptane, and CH_4/dimethoxymethane [100,101] mixtures. The added reactions can be found in the supplementary material. The mechanism of Cai and Pitsch was developed for describing the combustion behavior of gasoline surrogate fuel. It contains no DME chemistry and a lot of species which are not interesting for our conditions. Therefore, only its benzene and PAH chemistry was considered here. This chemistry up to 4-ring PAHs contains only 67 species and 162 reactions so that the PolyMech2.0 is still short but can now describe well acetylene and benzene concentrations and contains also pyrene which can be used as indication for soot formation. The prediction of soot formation trends by using the square of the pyrene concentration [102] is very important for the use of the mechanism for simulating technical polygeneration processes.

In addition, the rate constant for the sensitive reaction of propyne and H radicals to C_2H_2 and CH_3 was adapted from Wang et al. [93]. Besides the reactions indicated by the sensitivity analyses, the rate constant for the thermal decomposition of C_2H_4 to C_2H_2 and H_2 (reaction 13) was changed to the values integrated in the GRI 3.0 mechanism [86]. This change was required, because the rates obtained by the PolyMech1.0 expression were too high at temperatures higher than 1000 K, compared to literature values, and led to a different temperature dependence of the C_2H_4 mole fractions compared to the shock-tube experiments. The rate expression is also used in the AramcoMech3.0 yielding H_2CC instead of C_2H_2 directly. But the rates for the isomerization of H_2CC to C_2H_2 are very high, thus the addition of this reaction did not change model predictions and was therefore omitted.

Using the PolyMech1.0, a much lower propane (C_3H_8) conversion was observed compared to the plug-flow reactor experiments, especially at $\phi \geq 10$. The sensitivity analysis revealed the H-abstraction by OH forming iso- and *n*-C_3H_7 as two of the most sensitive reactions. These rate expressions were updated by the more recent ones proposed by Sivaramakrishnan et al. [92].

The addition of several reactions describing the oxidation of C_2H_6 and C_3H_8 did not significantly improve the model predictions and was not considered in order to keep the final modified version of the PolyMech as small as possible. The new rate coefficient of the methane decomposition to methyl and H atoms of Wang et al. [94] has also almost no influence on the present results but the old rate coefficient was replaced because the new value improves the simulation of methane pyrolysis at high temperatures significantly [103].

3.2. Shock-tube modeling

Ignition delay times and species mole fractions were simulated with Chemical Workbench® [104] using the conP approach (adiabatic isentropic compression or expansion) based on predefined pressure profiles that are derived from experiments with non-reactive test-gas mixtures. Default values (relative tolerance: 10^{-5}, absolute tolerance: 10^{-16}) were used for the tolerances. For measurements at $\phi = 2$, a pressure increase of 2 %/ms was considered for the initial 2.8 ms followed by a constant pressure. At $\phi = 10$, no pressure increase was observed, thus the simulations were performed with the initial pressure p_5. For the simulations of the species mole fraction profiles at $\phi = 10$, a constant pressure was used for the initial 12 ms followed by a pressure decrease of 12 % of p_5/ms (caused by the cooling due to rarefaction waves) for 7 ms (based on an averaged sampling time) and a very fast pressure decrease to 50 mbar (by expansion into the evacuated sampling vessel). The simulation is continued at this low pressure/temperature for 6 ms. The calculated species mole fractions are then normalized relative to the calculated Ne mole fraction at this time ($x_{i,n} = x_i \times 2\ \%/x_{Ne}$, 2% Ne as internal standard in the initial mixture) to compare them to the standard-normalized measured values. Sensitivity analyses were performed with the conP approach and a constant pressure for 15 ms. The calculated product mole fractions for this reaction time are very similar to the ones calculated with cooling. The times chosen for the simulation of the products were selected for the following reasons: The concentrations of almost all species reach a quasi-constant level at about 2 ms after ignition (see Ref. [35] for a similar mixture). Only the concentrations of acetylene and especially benzene vary after these 2 ms. The increase of the benzene concentration could not be stopped in the cooling phase by rarefaction waves because the temperature decrease is too low due to the high heat capacity of the mixture and the radical chain reactions are continued. The reactions are then stopped by the expansion into the sampling vessel. The amount of gas in the reaction vessel is small compared to the amount in the shock tube so that the pressure and temperature in the shock tube are not influenced by the sampling. Simulating the reactions also in the cooling phase until 7 ms after the opening of the sampling valve gives a good approximation of the sampling, considering that the pressure is strongly decreasing during the sampling time of 10 ms. More details, such as calculated species concentrations and an experimental pressure profile are given in Ref. [35]. The ignition delay times at $\phi = 2$ are determined by Chemical Workbench from the time when CH reaches its maximum value, which is identical to the occurrence of the CH* chemiluminescence peak. Both peaks are so narrow that it makes no difference if the peak position or the interpolation of the steepest increase of the signal intensities to the zero line are used as indicator for the ignition delay time. The simulations at $\phi = 10$ show very broad CH* and OH* peaks. For this reason, ignition delay times were determined by extrapolating the steepest increase of the simulated CH* mole fraction to the zero line.

3.3. Plug-flow reactor modeling

For the flow reactor simulations, the plug-flow model with fixed temperature profile of ChemKin Pro 19.0 [105] was used. Besides the geometric specification of the reactor, the complete temperature profiles, including the heating and cooling zone at the in- and outlet are used for the simulations. The temperature profiles were measured prior to the experiments using a moving type-K thermocouple along the center axis of the reactor in an inert gas flow of argon. Temperature profiles, inlet conditions and measured mole fractions are provided as SM.

To estimate the temperature rise of the gas mixture due to exothermal reactions, additional simulations were performed for the mixture with the highest heat release ($\phi = 2$, 10 % DME). For this simulation, the shear-flow model of ChemKin Pro 19.0

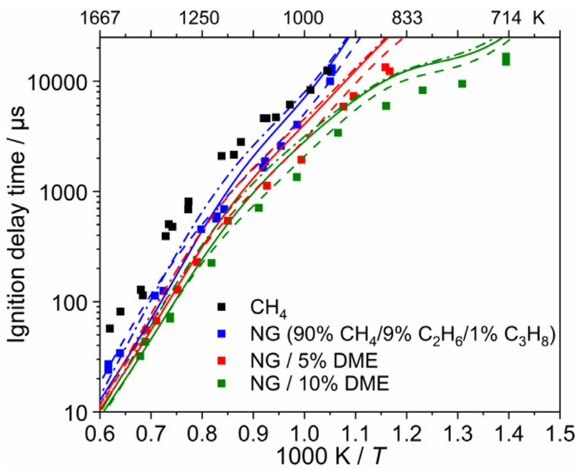

Fig. 4. Measured and simulated ignition delay times of natural gas/DME/air mixtures at $\phi = 2$ and 30 bar. Symbols: Experiments, full lines: Simulations with PolyMech1.0 [35], dash-dotted lines: Simulations with PolyMech2.0, dashed lines: Simulations with AramcoMech3.0 [42].

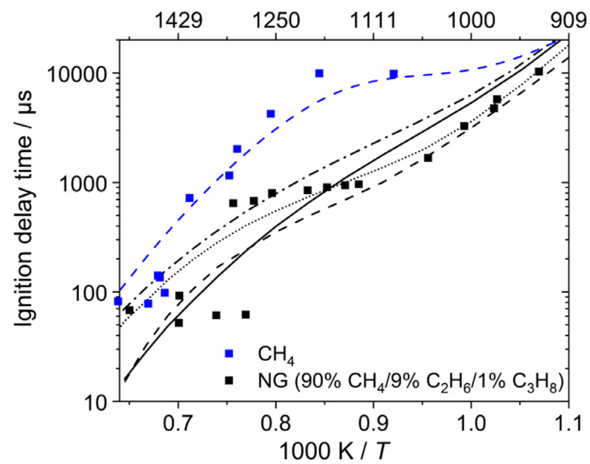

Fig. 5. Measured and simulated ignition delay times of CH_4 and a natural gas/air mixture at $\phi = 10$ and 30 bar. Symbols: Experiments, full lines: Simulations with PolyMech1.0 [35], dash-dotted lines: Simulations with PolyMech2.0, dashed lines: Simulations with AramcoMech3.0 [42], dotted lines: Simulations with the Cai and Pitsch [99] mechanism.

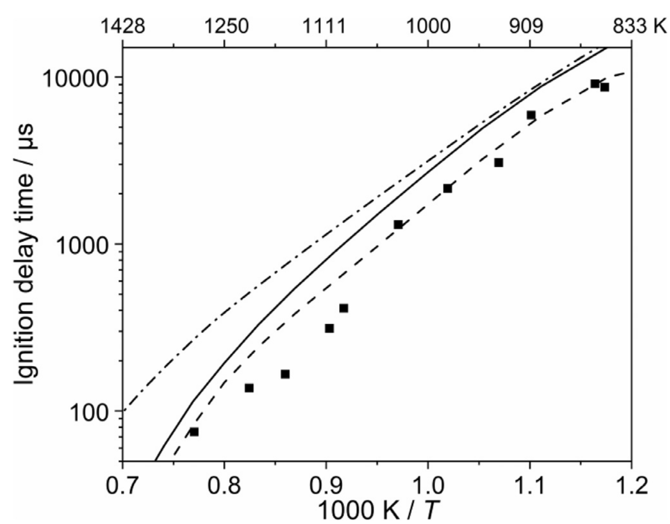

Fig. 6. Measured and simulated ignition delay times of a natural gas/5 % DME/air mixture at $\phi = 10$ and 30 bar. Symbols: Experiments, full lines: Simulations with PolyMech1.0 [35], dash-dotted lines: Simulations with PolyMech2.0, dashed lines: Simulations with AramcoMech3.0 [42].

was used, taking radial diffusion, the heat transfer from the wall to the gas and the exothermal heat release into account. It is a two-dimensional approach, predicting temperature, velocity and species concentration fields within the reactor.

4. Results and discussion

4.1. Shock-tube

Ignition delay times ($\phi = 2$ and 10) and product composition ($\phi = 10$) of natural gas and natural gas/DME mixtures were determined in a high-pressure shock tube and compared to predictions using the AramcoMech3.0 [42], PolyMech1.0 [35], and PolyMech2.0 (see section 3.1).

4.1.1. Ignition delay times

Measured and simulated IDTs at $\phi = 2$ are presented in Fig. 4. The data with natural gas as fuel show shorter IDTs compared to neat methane data [101]. A further reduction of the IDTs is achieved by the addition of DME. The AramcoMech3.0 [42] predicts all measured IDTs very well. Only for the experiments with 10% DME addition at temperatures $<$ 900 K about 50 % longer IDTs are predicted. The simulations with the PolyMech1.0 [35] and the measured IDTs for natural gas and natural gas/DME mixtures agree very well for temperatures above 1200 K. For lower temperatures, 50–90 % too long ignition delay times are predicted. The PolyMech2.0 predicts about 25 % longer IDTs at higher temperatures compared to the original version whereas at the lower temperature end the predictions are almost identical.

Measured and simulated IDTs of natural gas/air mixtures at $\phi = 10$ are shown in Fig. 5. For temperatures above 1200 K it was very difficult to determine the IDTs because the OH* and CH* peaks exhibit two distinct steep increases and very broad peaks. This behavior has not been observed before for CH_4, CH_4/additive and natural gas/DMM mixtures at $\phi = 10$ [100,101] and is also not predicted by the simulations. The measured OH* and CH* chemiluminescence signals and the corresponding simulations using the AramcoMech3.0 can be found in the supplementary material at temperatures of 974, 1204, and 1427 K. A pressure increase by the ignition could not be observed. We assume that the first increase is caused by ethane/propane oxidation and the second by methane oxidation. The IDTs of natural gas are much shorter compared to methane and are similar to the values at $\phi = 2$. The measured IDTs are very well predicted by the AramcoMech3.0 [42] and the Cai and Pitsch [99] mechanism. For lower temperatures, about 50 % too long IDTs are predicted by the PolyMech1.0 [35]. Especially at higher temperatures the modifications of the PolyMech1.0 cause longer ignition delay times. The same effect was observed for the natural gas/5% DME mixture at $\phi = 10$ (Fig. 6). It is mainly caused by the lower rate coefficient for reaction 13 ($C_2H_4(+M) = C_2H_2 + H_2(+M)$), so that less H_2, which accelerates the ignition, is produced.

Measured and simulated IDTs of a natural gas/5 % DME/air mixture at $\phi = 10$ are shown in Fig. 6. The IDTs are reduced by the DME addition. Measurements and simulations with the AramcoMech3.0 [42] agree very well whereas the PolyMech1.0 [35] predicts about 50 % too long IDTs.

4.1.2. Product formation

The product composition of a natural gas/air mixture at $\phi = 10$ after ignition, which is very similar to experiments with CH_4, CH_4/n-C_7H_{16}, CH_4/diethyl ether [101], CH_4/DME [35] and CH_4/dimethoxymethane [100] as fuel, is shown in Fig. 7. Main

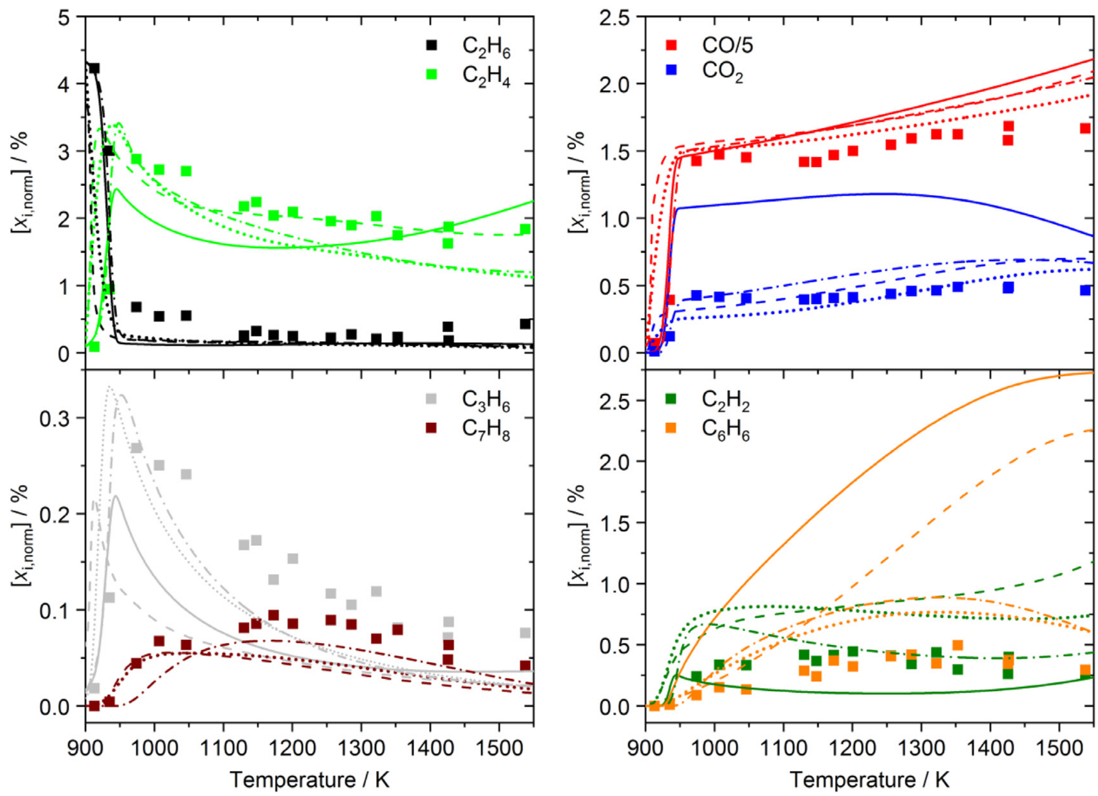

Fig. 7. Measured and simulated species mole fractions of a natural gas/air mixture at $\phi = 10$ and 30 bar after ignition. Symbols: Experiments, full lines: Simulations with PolyMech1.0 [35], dash-dotted lines: Simulations with PolyMech2.0, dashed lines: Simulations with AramcoMech3.0 [42], dotted lines: Simulations with the Cai and Pitsch mechanism [99].

products are CO, H_2 (not quantified), CO_2, H_2O (not quantified), C_2H_2, C_2H_4, C_2H_6, C_3H_6, benzene and toluene. About half of the consumed fuel is converted to CO, the other half to hydrocarbons. The experimental results were compared to the predictions of four mechanisms: Cai and Pitsch [99], AramcoMech3.0 [42], PolyMech1.0 [35], and the PolyMech2.0 (section 3.1). The simulations with the Cai and Pitsch mechanism [99] agree very well for all products with the exception of C_2H_2, benzene (temperature dependence is very well predicted but measured values are 40 % lower than the simulated ones) and C_2H_4 (measured values are 30 % higher than the simulated ones except at low temperatures). The AramcoMech3.0 [42] predicts all product mole fractions very well with the exception of much too high mole fractions of acetylene and benzene (up to a factor of eight). Contrary to the Cai and Pitsch [99] mechanism, the benzene mole fraction constantly increases with higher temperatures in the experiments. The benzene consumption reactions to form higher PAHs contained in the mechanism are obviously too slow. The original PolyMech1.0 [35] predicts only the CO and C_3H_6 mole fractions well. Much too high mole fractions for CO_2 (about a factor of two) and benzene (up to a factor of ten) and much too low mole fractions for C_2H_2 (about a factor of four) are predicted. At low temperatures, too low C_2H_4 mole fractions are predicted whereas at high temperatures a good agreement for C_2H_4 is observed. With the PolyMech2.0, the predictions of all these species mole fractions are significantly improved, the simulations for benzene and C_2H_4 are now very similar to the ones of the Cai and Pitsch mechanism [99], the predicted CO_2 mole fractions are reduced, and C_2H_2 is now predicted very well. Toluene is now included in the mechanism and is also predicted very well. The integration of the benzene and PAH chemistry of the Cai and Pitsch mechanism [99] was successful, the slight differences in the predicted benzene mole fractions are caused by different mole fractions of species forming benzene. With the PolyMech2.0, a mechanism with relative few species and reactions is now available which can predict all observed product species well.

The product composition of a natural gas/5 % DME/air mixture at $\phi = 10$ after ignition is shown in Fig. 8. The measured and simulated product composition is almost identical to the experiments without DME as additive but is shifted to about 100 K lower temperatures due to the shorter ignition delay times. The results for the comparison of the simulations with the different mechanisms and the experiments are identical to the ones without DME but simulations with the Cai and Pitsch mechanism [99] were not possible because it contains no DME submechanism.

4.2. Plug-flow reactor

In this section, experimental species profiles of the fuel and major products as a function of temperature and equivalence ratio will be presented and compared to model predictions and results of neat natural gas/O_2 experiments published earlier [20]. In addition, reaction pathways towards interesting products are analyzed to assess the influence of DME on the partial oxidation process.

The TOF-MS signals of other detected species relative to the argon signal can be found in the supplementary material. Although these species mole fractions were not quantified, the trends as a function of temperature can be used for validation purposes. The results of the shear-flow simulation mentioned in Section 3.3 showed a maximum temperature rise of ~ 84 K at a nominal temperature of 873 K within the isothermal zone. Nevertheless, the difference between the calculated mole fractions of the shear-flow and plug-flow simulations at the end of the reactor was less than 10 % (see supplementary material). A negligible difference in calculated mole fractions was observed for lower and

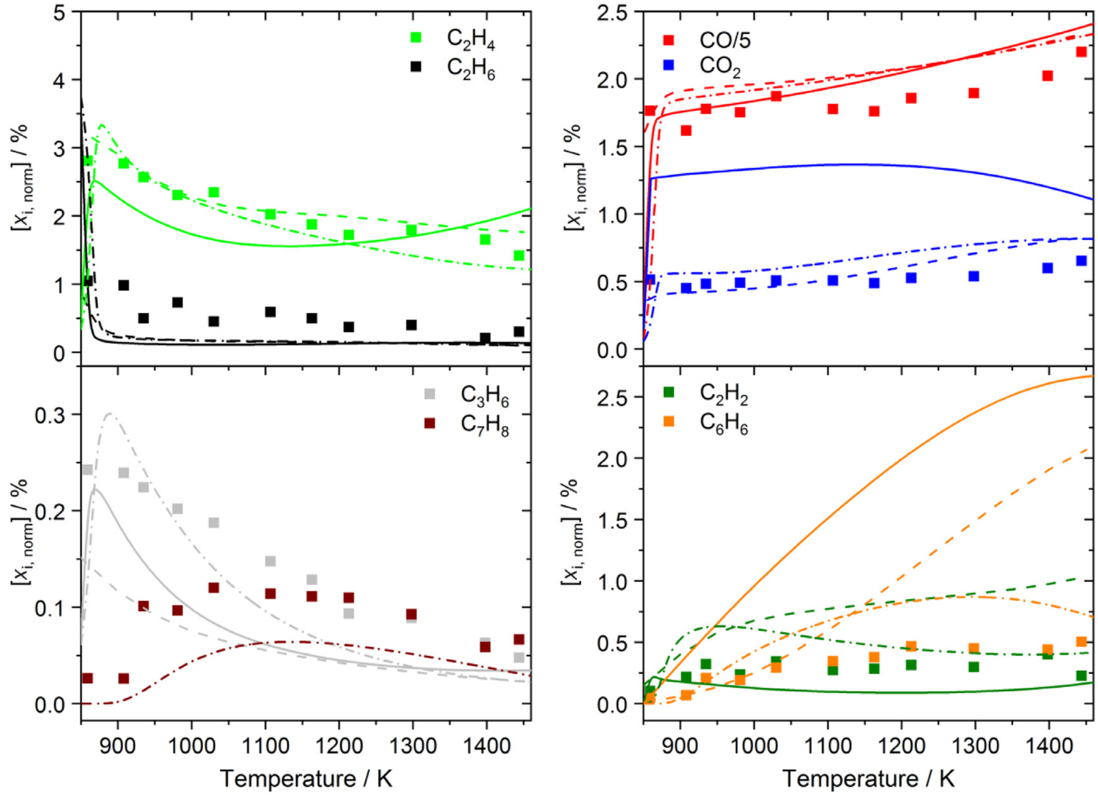

Fig. 8. Measured and simulated mole fractions of a natural gas/5 % DME/air mixture at $\phi = 10$ and 30 bar after ignition. Symbols: Experiments, full lines: Simulations with PolyMech1.0 [35], dash-dotted lines: Simulations with PolyMech2.0, dashed lines: Simulations with AramcoMech3.0 [42].

higher temperatures so that it seems justified to present only the results of plug-flow simulations in this section.

4.2.1. Comparison of experimental and modeling results

Figures 9 and 10 show the mole fractions of major species as a function of temperature and equivalence ratio for the 5 and 10 % DME mixtures, respectively. Experimental results (symbols) are compared to simulations (lines).

4.2.1.1. Influence of DME on fuel conversion. As seen in Figs. 9 and 10, the conversion of DME starts between 523 and 573 K in all natural gas/DME experiments, reaching a local maximum at 623 K, followed by a local minimum between 673 and 723 K. For the 5 % DME mixture at $\phi = 2$, only a relatively slow conversion up to a temperature of 723 K is observed. At higher temperatures, the conversion of DME increases for all mixtures until total depletion is reached at temperatures between 773 and 973 K, depending on the equivalence ratio. This behavior, which is more pronounced at high equivalence ratios as well as for fuel mixtures with a natural gas/DME ratio of 9/1, can be attributed to the negative temperature coefficient (NTC) behavior of DME: Similar results can be found for CH_4/DME mixtures presented in Refs. [34,35]. In Ref. [34] the NTC region is slightly shifted towards lower temperatures because of higher reactor pressures. The high reactivity of DME and the associated formation of radicals at low temperatures initiate the conversion of all alkanes at 200 K lower temperatures compared to the single fuel experiments (neat natural gas) presented in Ref. [20]. The conversion of CH_4 stays low (< 5 %) at temperatures up to 773 K, whereas more than 20 and 40 % C_2H_6 and C_3H_8, respectively, are converted at the first NTC inflection point (623 K). It must be noted that CH_4 contains around 75 and 68 % of the total amount of carbon in the 5 and 10 % DME mixtures, respectively. This means that even if less CH_4 than C_2H_6 or C_3H_8 is converted it can contribute more carbon to the products. The conversion of CH_4 is counteracted by methane producing pathways from DME, C_2H_6, and C_3H_8. This effect is apparent in the reaction pathway analysis for the 10 % DME mixture at 973 K and $\phi = 20$ at the time corresponding to 25 % DME conversion in Fig. 11. It is clearly seen that the reactions of all reactants, mainly with CH_3 radicals, lead to the formation of CH_4. Also, the reaction of CH_2O with CH_3 contributes largely to CH_4 production. Although most of CH_3 is produced via H-abstraction from CH_4 by OH, the net carbon flux between these two species is towards CH_4. The formation of CH_4 exceeds its consumption under these conditions. The reaction onset temperature in natural gas mixtures is lowered which is desirable in polygeneration processes with auto-ignition at realistic compression ratios.

For C_3H_8, a second NTC region is predicted by the models for the $\phi \geq 10$ mixtures at temperatures around 750 K. Such an NTC behavior was also observed by Gallagher et al. [106], who experimentally investigated C_3H_8 oxidation at equivalence ratios between 0.5 and 2 using a rapid compression machine. Here, experimental evidence for the NTC region can be observed for the $\phi = 10$ mixtures at $T \approx 800$ K, whereas only a slight decrease in reactivity is visible in the $\phi = 20$ mixtures. To clearly resolve the NTC behavior in this narrow temperature range of about 100 K, more temperature steps would have been necessary in the experiments. The model predictions of all mechanisms for CH_4, C_2H_6, DME, and O_2 are in very good agreement with the experimental data for both conditions in Figs. 9 and 10, but the PolyMech2.0 shows a slightly better agreement at higher equivalence ratios. In contrast, C_3H_8 is overpredicted at temperatures higher than 773 K for $\phi \geq 10$ by all three models. While the highest deviation is shown by the PolyMech1.0 for $\phi \geq 10$, the AramcoMech3.0 shows the best overall agreement. This behavior can probably be attributed to the very detailed C_3-subset, including peroxide chemistry. The best agreement for $\phi = 2$ is observed for the

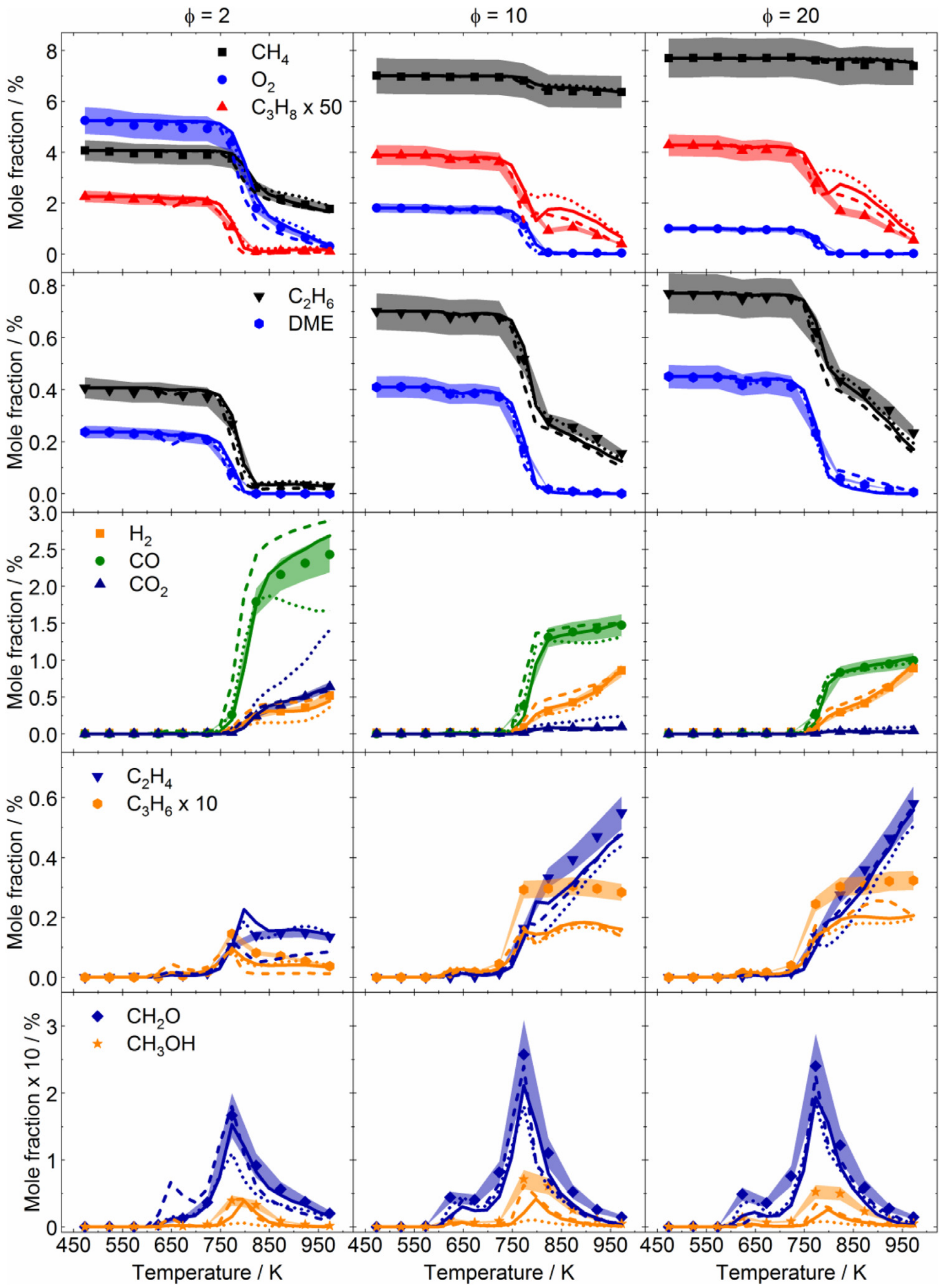

Fig. 9. Mole fractions of reactants and products as a function of temperature for different equivalence ratios for the 95/5 natural gas/DME mixture. Symbols: TOF-MS experiment, lines: Simulations based on the PolyMech1.0 [35] (dotted lines), AramcoMech3.0 [42] (dashed lines), and the PolyMech2.0 (solid lines). Mole fraction uncertainties are displayed by the shaded areas.

PolyMech2.0 together with an improvement at all equivalence ratios compared to the unmodified version.

To sum up, DME leads to radical formation at low temperatures and initiates the conversion of the alkanes in the mixture at 200 K lower temperatures compared to neat natural gas/O_2 mixtures shown in Ref. [20]. In addition, the conversion of DME, but also the conversion of ethane and propane lead to CH_4 production and even higher CH_4 mole fractions at $\phi = 10$ compared to the initial amount in the mixture. The PolyMech2.0 shows better predictions of the mole fractions of all reactants compared to the unmodified mechanism. All three reaction mechanisms show a good agreement to the experimental data.

Regarding product formation, synthesis gas (H_2/CO), unsaturated hydrocarbons (C_2H_4, C_3H_6), oxygenated species (CH_2O, CH_3OH), CO_2 and H_2O (not shown here, see supplementary material) are found as major products and will be discussed in the following.

4.2.1.2. Synthesis gas and CO_2. CO and CO_2 are preferentially formed at low ϕ and high temperatures with the CO_2 mole fractions becoming negligibly small at $\phi \geq 10$. In contrast, CO remains the main product throughout the investigated mixtures with maximum mole fractions of ~ 2.5 % and ~ 1 %, going from $\phi = 2$ to $\phi = 20$, independent of the DME amount in the mixture. The mole fraction of H_2 increases with increasing temperature, equivalence ratio and DME amount, reaching 0.5 % at $\phi = 2$ (natural gas/DME = 95/5) and 1.2 % at $\phi = 20$ (natural gas/DME = 90/10) and the highest investigated temperature of 973 K, respectively. In

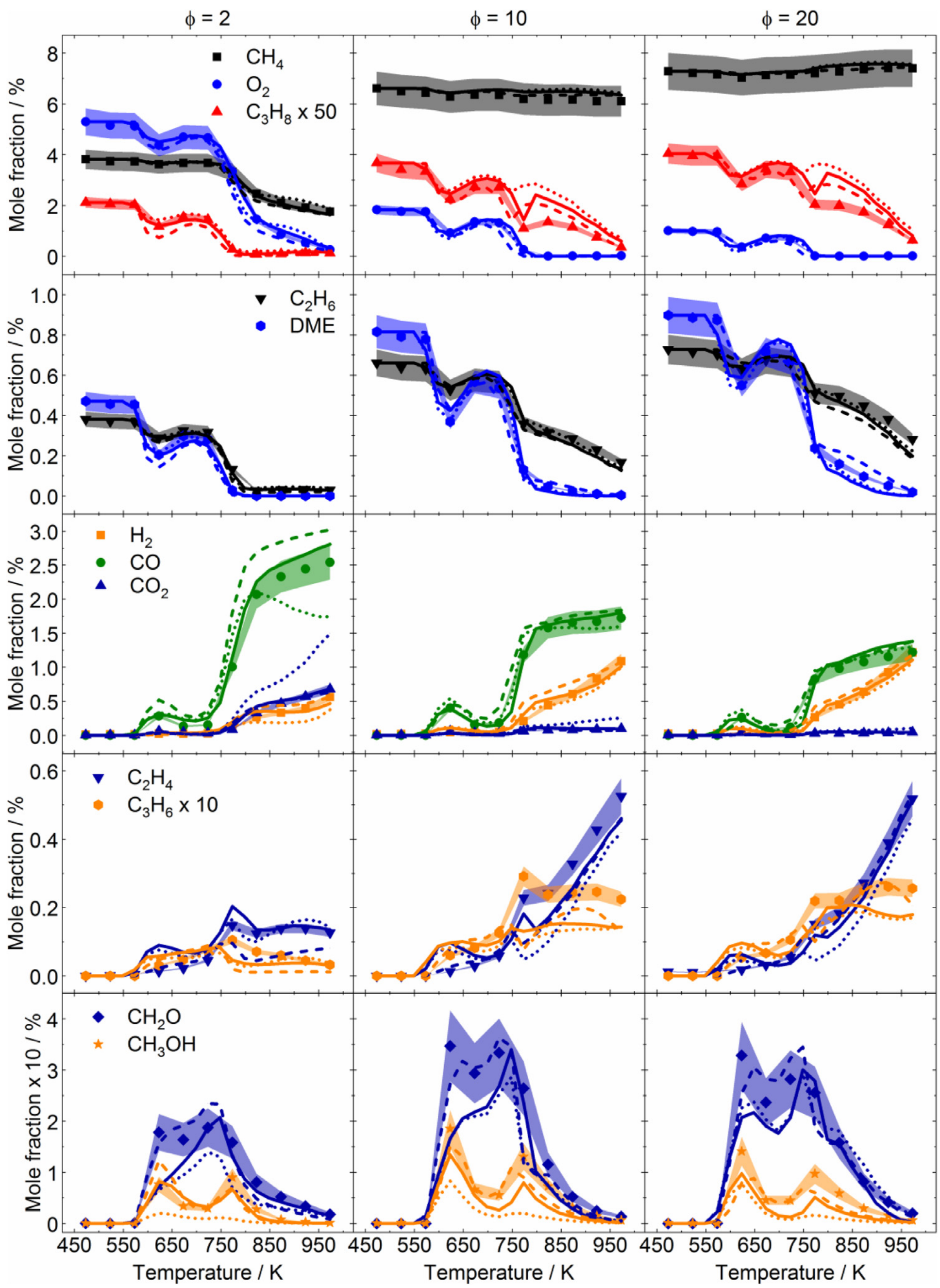

Fig. 10. Mole fractions of reactants and products as a function of temperature for different equivalence ratios for the 90/10 natural gas/DME mixture. Symbols: TOF-MS experiment, lines: Simulations based on the PolyMech1.0 [35] (dotted lines), AramcoMech3.0 [42] (dashed lines), and the PolyMech2.0 (solid lines). Mole fraction uncertainties are displayed by the shaded areas.

addition, the NTC region of DME leads to a local maximum in the species mole fractions at 623 K, especially visible for CO and the 10 % DME mixtures (Fig. 10). With respect to these species, the PolyMech2.0 shows the best agreement with the experiments. While the AramcoMech3.0 overpredicts the synthesis gas mole fractions at $\phi = 2$, the PolyMech1.0 underpredicts synthesis gas at $\phi = 2$ and $T > 800$ K and extremely overpredicts CO_2 for all mixtures at $T > 800$ K.

4.2.1.3. Unsaturated hydrocarbons. The formation of the unsaturated hydrocarbons C_2H_4 and C_3H_6 is favored at high equivalence ratios and high temperatures, as it is shown in Figs. 9 and 10. At $\phi = 2$ the mole fraction of C_2H_4 nearly remains constant at about 0.2 % at temperatures higher than 800 K. It increases with increasing temperature for the mixtures at $\phi = 10$ and 20 to more than 0.5 %. Figure 11 (10 % DME mixture at $\phi = 20$ and 973 K) and Fig. 12 (5 % DME mixture at $\phi = 10$ and 773 K) reveal that most C_2H_4 is formed from C_2H_6 and C_3H_8. At $\phi = 2$, the higher amount of oxygen and the associated higher reactivity of the mixture lead to a depletion of C_2H_6 and C_3H_8 for $T > 800$ K, explaining the observed temperature dependence of the C_2H_4 mole fraction. The lack of oxygen plays a key role in the formation of the unsaturated hydrocarbons. Reaction path analyses performed for the mixtures at $\phi \geq 10$ and temperatures above 900 K with the PolyMech2.0 reveal a slow but steady increase in the importance of pyrolysis reactions. For example, at temperatures higher than 800 K the long reaction times after the consumption of oxygen in the first 20 cm of the reactor, can lead to the pyrolysis of C_2H_6 and C_3H_8 as well as C_3H_6. Typical

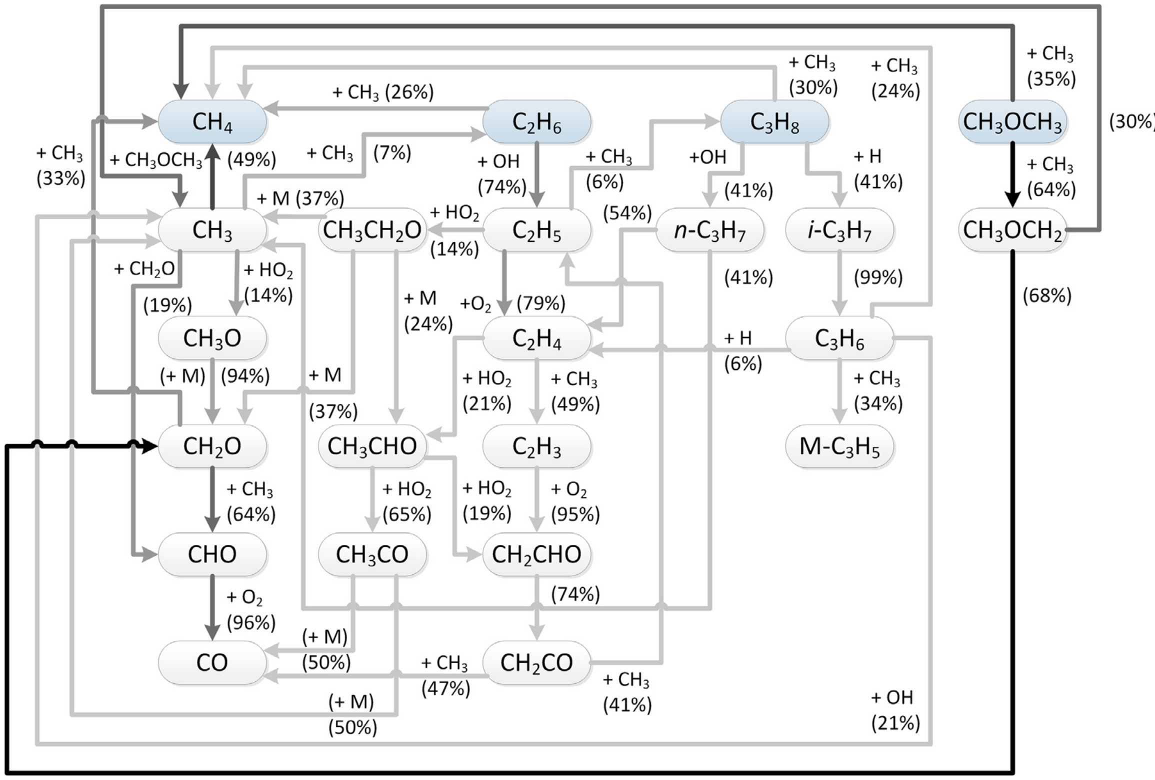

Fig. 11. Reaction path analysis using the PolyMech2.0 for the 10 % DME mixture at 973 K and ϕ = 20 (calculated at the time corresponding to 25 % DME conversion). The percentages show the relative rate of carbon consumption of the limiting reactant and line darkness represents the percentage of carbon flux, related to the highest carbon flux within the reaction path, which is found between CH_3OCH_3 and CH_3OCH_2. Species next to the arrows represent the main reaction partners.

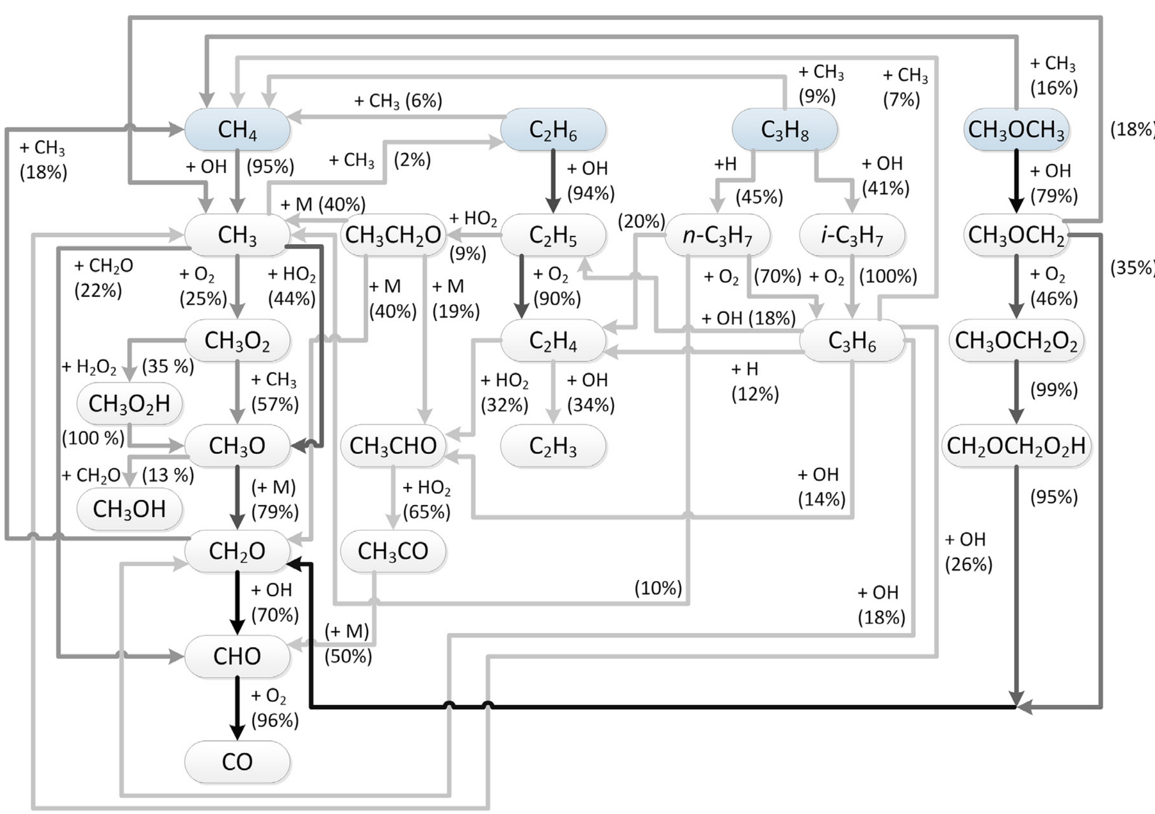

Fig. 12. Reaction path analysis using the PolyMech2.0 for the 5 % DME mixture at 773 K and ϕ = 10 (calculated at the time corresponding to 25 % DME conversion). The percentages show the relative rate of carbon consumption of the limiting reactant and line darkness represents the percentage of carbon flux, related to the highest carbon flux within the reaction path, which is found between CH_2O and CHO. Species next to the arrows represent the main reaction partner.

products are C_3H_6, C_2H_4, H_2 and CH_4 [107,108] in the $\phi \geq 10$ mixtures.

The simulations using the PolyMech2.0 show the best agreement with the experiments for the 5 % DME mixtures at all temperatures and the 10 % DME mixtures at temperatures above 700 K, both qualitatively and quantitively. The unmodified version predicts slightly lower mole fractions at $\phi \geq 10$. The AramcoMech3.0, however, underpredicts C_2H_4 mole fractions at $\phi = 2$ and $T > 800$ K and at the same time overpredicts CO at these conditions. This is partly due to lower reaction rates for the recombination of CH_3 radicals to C_2H_6 in the AramcoMech3.0 compared to the PolyMech2.0, so that the oxidation pathway ($CH_3 \rightarrow CO$) is favored compared to the recombination pathway yielding C_2H_6 and subsequently C_2H_4. At higher ϕ, the oxidation pathway becomes less important (see Figs. 11 and 12), which leads to a better agreement with the experiments.

In contrast to C_2H_4, the C_3H_6 mole fractions start to decrease at temperatures above 800 K for the $\phi = 2$ mixtures. C_3H_8 is completely consumed at these conditions and O_2 is still available for the reaction to proceed so there are two options for C_3H_6 to be consumed and expected to occur simultaneously: The pyrolysis of C_3H_6 and, since C_3H_6 is more reactive than C_2H_4, the scavenging of available radicals. At higher equivalence ratios, oxygen is depleted at $T > 800$ K so that fewer radicals are available. The missing radicals and probably the competition between pyrolysis of C_3H_6 and C_3H_8, which is still available at $\phi > 2$, contribute to nearly constant mole fractions of C_3H_6. According to literature, C_3H_6 is not the main product during the pyrolysis of C_3H_8, but C_2H_4 and CH_4 [107]. Nevertheless, a small part of C_3H_8 will undergo H-abstraction by H radicals yielding propyl radicals and finally form C_3H_6. The dominant channel of C_3H_6 decomposition is the pyrolysis to allyl (C_3H_5) and H radicals [109]. While H radicals can react with C_3H_6 to form $H_2 + C_3H_5$, $CH_3 + C_2H_4$ or $C_3H_5 + CH_4$, for example, the allyl radicals can also lead to the formation of higher hydrocarbons like 1,3-butadiene (C_4H_6), cyclopentadiene (C_5H_6) or benzene (C_6H_6). These species were observed in flow reactor experiments, presented by Wang et al. [108]. All of these hydrocarbons were detected in the TOF-MS experiments at $\phi > 2$ in small quantities. Signals of fragments formed during the ionization of these C_4 and C_5 species overlap with the signal of C_3H_6 and can contribute to the discrepancy between simulation and measured mole fraction in the experiments with $\phi > 2$ above 700 K. But this experimental uncertainty alone cannot explain the large differences. This conclusion is supported by the fact that a reasonable agreement between experimental and simulation results, obtained with the AramcoMech3.0, is found for the $\phi = 20$ mixtures, at which the highest signals for the higher hydrocarbon species were detected. Missing reactions or outdated rate expressions need to be considered with respect to the competition between oxidation and pyrolysis of C_3H_8 and C_3H_6.

In summary, the unsaturated hydrocarbons C_2H_4 and C_3H_6 are preferentially formed at large equivalence ratios and high temperatures. The main reaction pathways to these species are found to be reactions of different radicals with the respective alkane (C_2H_6 and C_3H_8) and at high temperatures ($T > 800$ K) also their pyrolysis. The modified reaction mechanism predicts the mole fractions of C_2H_4 well, but missing reactions are suggested to be the reason for poor predictions of C_3H_6 mole fractions compared to the very detailed AramcoMech3.0.

4.2.1.4. Oxygenated species. The mole fraction profiles of CH_2O and CH_3OH show two local maxima at 623 K and 723 – 773 K (Figs. 9 and 10). In general, the formation of both species is favored at high equivalence ratios with maximum mole fractions of 0.26 % (CH_2O) and 0.07 % (CH_3OH) for the 5 % DME mixture and 0.35 % (CH_2O) and 0.18 % (CH_3OH) for the 10 % DME mixture at $\phi = 10$. Because the mole fractions are much higher compared to the neat natural gas experiments, presented in Ref. [20] (0.07 % CH_2O and 0.01 % CH_3OH), both species are suggested to be formed mainly from DME oxidation. The maximum of both species is shifted to 100 K lower temperatures, compared to the neat natural gas mixtures [20].

With respect to CH_2O, both the AramcoMech3.0 and the PolyMech2.0 predict the mole fractions of both species in good agreement with the experiments. Deviations occur for the first maximum between 623 K and 723 K for the 5 % DME mixture at $\phi = 2$ where the AramcoMech3.0 overpredicts the mole fractions and between 573 K and 723 K for all 10 % DME mixtures where the PolyMech2.0 underpredicts the mole fractions. This overprediction of the mole fraction of CH_2O by the AramcoMech3.0 coincides with a much higher conversion of DME as well as C_2H_6 and C_3H_8 and overprediction of the mole fractions of C_2H_4, C_3H_6, and CH_3OH, compared to the PolyMech2.0. In case of the AramcoMech3.0 the reactivity of DME is predicted to be too high, leading to the formation of radicals, the subsequent conversion of C_2H_6 and C_3H_8, and finally the production of CH_2O. Regarding the PolyMech2.0, the predicted lower CH_2O mole fractions correspond to higher mole fractions of C_2H_4, compared to the experiment and the predicted mole fractions using the AramcoMech3.0. A reaction path analysis with the AramcoMech3.0 at $\phi = 2$ and 623 K for the 10 % DME mixture reveals three different pathways for CH_2O formation: 1) via the C_2-peroxide pathway and the sequence: $C_2H_6 \rightarrow C_2H_5 \rightarrow C_2H_5O_2 \rightarrow C_2H_5O_2H \rightarrow C_2H_5O \rightarrow CH_2O$, 2) via the C_1-peroxide pathway and the same sequence as before starting with CH_4 and 3) via decomposition of $CH_3OCH_2O_2$ and $CH_2OCH_2O_2H$. Both the C_2-peroxide pathway, in which the decomposition of $C_2H_5O_2H$ releases OH radicals, and the decomposition of $CH_3OCH_2O_2$ yielding two CH_2O molecules and an OH radical, are not included in the PolyMech. The extra OH radicals further enhances CH_4 consumption and thus promotes pathway 2, also increasing the formation of CH_2O. In addition, the C_2-peroxide pathway leads to smaller C_2H_4 mole fractions, as the rate of the reaction $C_2H_5 + O_2 = C_2H_5O_2$ is much higher (two orders of magnitude) compared to the rate of the reaction $C_2H_5 + O_2 = C_2H_4 + HO_2$, which is the main pathway of C_2H_5 consumption in case of the PolyMech under these conditions. These differences explain the deviations in the predicted C_2H_4 and CH_2O mole fractions by both mechanisms. The rather complex C_2-peroxide chemistry and further DME reactions were not added to the PolyMech. This would have made the mechanism much larger and would only lead to a slightly better agreement in a small temperature range in the best case. The possibility of adding C_2- and C_3- peroxide chemistry in a reduced form without neglecting important pathways will be investigated in the future.

CH_3OH is also predicted well by the AramcoMech3.0 and PolyMech2.0, especially regarding the trend as a function of temperature. The remaining discrepancies show that the interaction between the natural gas components and DME has not yet been fully understood. But it can be suggested that the DME subset is responsible for the deviations at low temperatures, since it was shown by sensitivity analyses that the CH_3OH mole fractions are very sensitive to DME involved reactions (see Fig. 2).

To unravel the influence of DME on the production of both oxygenated species, a reaction pathway analysis was performed with the PolyMech2.0 at $\phi = 10$ and 773 K for the 5 % DME mixture, see Fig. 12.

A large part of CH_2O is formed by DME oxidation, more specifically the dissociation of CH_3OCH_2 and $CH_2OCH_2O_2H$. While the first species is formed by H-abstraction from DME, the second species is formed by isomerization of $CH_3OCH_2O_2$, produced by O_2 addition to CH_3OCH_2. Under these conditions, DME contributes to more than 58 % to the production of CH_2O by these two reactions. The remaining CH_2O is formed by the dissociation of CH_3O radicals, which in turn are nearly exclusively formed by reactions involving CH_3 radicals and the methyl peroxy radical. The highest carbon flux to CH_3 comes from CH_4, followed by CH_3OCH_2. In addition, pathways from CH_2O, C_2H_6, C_3H_8, and DME to CH_4 are found under these conditions. Finally, indirect reaction pathways from DME, C_2H_6, and C_3H_8 also lead to the formation of CH_2O, decreasing the contribution of CH_4.

For CH_3OH, a slightly different situation is observed, as it is nearly exclusively formed by reactions involving CH_3O radicals with a contribution of about 60 %. Although CH_3O contains carbon that comes from DME and the other educts via the reaction sequence (Educt → CH_4 → CH_3 → CH_3O), no important direct pathways to CH_3O and CH_3OH are observed under these conditions. So, the contribution of CH_4 to the formation of CH_3OH is expected to be higher compared to the formation of CH_2O. C_2H_6 and C_3H_8 contribute slightly to methanol production, which is seen by the light grey arrows, connecting their oxidation pathways with the one of CH_4.

With increasing DME amount in the mixture, the contribution of DME to the formation of both oxygenated species increases, as the direct pathways towards CH_2O (decomposition of CH_3OCH_2 and $CH_2OCH_2O_2H$) and CH_3OH (the reaction of DME and CH_3O radicals) are favored. Additionally, a larger part of CH_3 radicals is produced from the decomposition of CH_3OCH_2, further increasing the contribution of DME to CH_2O and CH_3OH production. Surprisingly, for the 10 % DME mixtures at $T = 623$ K, at which the first maximum in the CH_2O and CH_3OH mole fractions is observed, the contribution of DME with respect to CH_2O and CH_3OH production is much lower compared to the product contribution at higher temperature. This is because a large amount of oxygen is still available, favoring the CH_3OCH_2 → $CH_3OCH_2O_2$ pathway compared to the CH_3OCH_2 → CH_3 and CH_3OCH_2 → CH_2O pathways. Also, $CH_2OCH_2O_2H$ does not decompose to CH_2O to the same extent as shown in Fig. 12 for the 5 % DME mixture at 773 K. Instead, O_2 is added to the species yielding $O_2CH_2OCH_2O_2H$, which decomposes to HO_2CH_2OCHO and OH radicals. An excerpt of the reaction pathway including subsequent steps is shown in Fig. 13. It can be observed that the main products from DME oxidation at this temperature are CO and CO_2 instead of CH_2O and CH_3 or CH_4. Moreover, the higher amount of released OH radicals (shown in blue) lead to a CH_4 conversion of ~5 % and thus favor the pathways towards CH_3OH and CH_2O. Finally, a higher DME amount in the mixture enhances the formation of oxygenated species from CH_4 at low temperatures.

Summarizing the main findings, the conversion of DME leads to an enhanced production of CH_2O and CH_3OH via direct and indirect reaction pathways. The major fraction of this species is formed from DME. With increasing DME amount in the mixture, the contribution of CH_4 with respect to CH_2O and CH_3OH formation also increases, initiated by reactions of CH_4 with radicals formed during DME conversion. This increase of CH_4 reactivity only applies to low temperatures. At these conditions, the decomposition of intermediate species of the DME oxidation leading to the formation of CH_2O and CH_3OH is rather slow and enough oxygen is available to favor O_2 addition reactions. These addition products form CO and CO_2 and not CH_2O and CH_3OH. The formed radicals convert CH_4 to CH_3 so that CH_2O and CH_3OH are produced mainly from CH_4 at these conditions. Both the AramcoMech3.0 and the PolyMech2.0 shows a much better agreement to the experimental data than the PolyMech1.0.

4.2.2. Maximum yields of major products

In order to compare the product spectra of all mixtures with each other, yields are calculated according to Eq. 2 and shown in Fig. 14. In addition to the results of the natural gas/DME

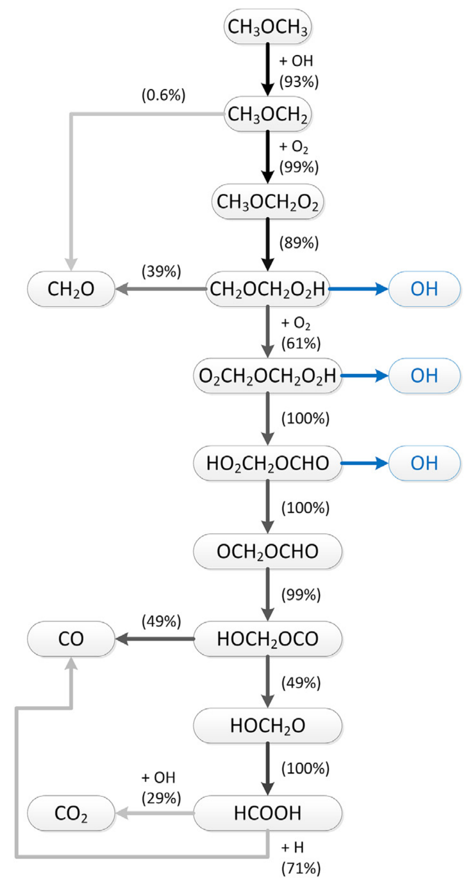

Fig. 13. Excerpt of the reaction path analysis using the PolyMech2.0 for the 10 % DME mixture at 623 K and $\phi = 10$ (calculated at the time corresponding to 25 % DME conversion). The percentages show the relative rate of carbon consumption of the limiting reactant and line darkness represents the percentage of carbon flux, related to the highest carbon flux within the reaction path, which is found between CH_3OCH_3 and CH_3OCH_2. Species next to the arrows represent the main reaction partner.

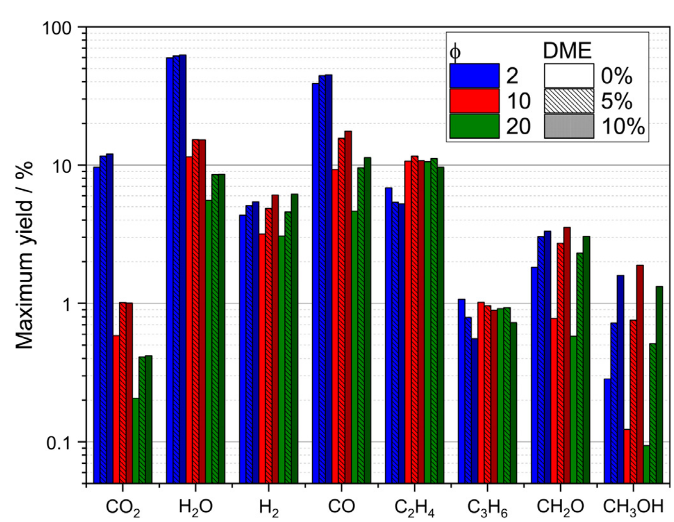

Fig. 14. Maximum yields of products as a function of equivalence ratio and DME amount. Values for neat natural gas (0 % DME) were taken from Ref. [20]. Yields do not add up to 100 % due to unconverted methane.

mixtures, the results of the neat natural gas/O_2 mixtures are shown for comparison.

$$Yield_i = \frac{x_i \times \text{number of C/H atoms in product species } i}{\sum_{\text{Reactants}} x_j \times \text{number of C/H atoms in reactant species } j} \quad (2)$$

As expected, the maximum yields of the common combustion products CO_2 and H_2O decrease rapidly with increasing equivalence ratio and slightly increase with the addition of DME. Higher yields of both H_2 and CO can be observed, if DME is added to the natural gas/O_2 mixture. While the yields of H_2 are nearly independent of the equivalence ratio, the maximum yields of CO decrease with increasing equivalence ratio due to the lower amount of oxygen. This results in an increasing H_2/CO ratio, making it more interesting for subsequent processes such as the Fischer-Tropsch synthesis. Despite lower yields compared to the DME mixtures, the highest H_2/CO ratio of ~1.3 is obtained for the neat natural gas/O_2 mixture at $\phi = 20$.

C_2H_4, mainly formed from C_2H_6 and C_3H_8 (see Figs. 11 and 12), is preferentially produced at high ϕ, because further reactions are prevented, if less O_2 is present in the mixture and fewer radicals are produced. In addition, no significant influence of DME on the formation of C_2H_4 is observed. The same is true for the yields of C_3H_6, except for $\phi = 2$, at which the higher reactivity of the DME mixtures is responsible for an enhanced C_3H_6 consumption, resulting in lower yields. The equivalence ratio does also not influence the C_3H_6 yields noticeably.

The last group of useful chemicals shown here are the oxygenated species represented by CH_2O and CH_3OH. While CH_2O and CH_3OH are important precursors for the production of several chemicals, CH_3OH is also an alternative transportation fuel. The formation of both species is significantly enhanced, if DME is added to the natural gas/O_2 mixtures, as seen in Fig. 14. Both species are easy to liquify so that they could be separated from the product gas at low cost.

5. Conclusions

The homogenous partial oxidation of natural gas/DME mixtures was investigated experimentally using a plug-flow reactor and a high-pressure shock tube at relevant conditions for polygeneration processes. The experiments cover equivalence ratios of 2, 10, and 20, temperatures between 473 and 1400 K, and pressures of 6 and 30 bar. In addition, the amount of DME was varied to assess its influence on the oxidation process. The addition of DME effectively reduces the reaction onset temperature of the natural gas/O_2 mixture by 200 K compared to neat natural gas/O_2 and promotes the formation of several useful products like synthesis gas and the oxygenated species CH_2O and CH_3OH. At the highest investigated equivalence ratio of 20, the replacement of 10 % fuel by DME lead to CH_4 production instead of consumption due to additional CH_4-producing pathways from DME. This observation was discussed by reaction path analyses, using PolyMech2.0, a modified version of the mechanism of Porras et al. [35]. To overcome this challenge, ozone could be a viable alternative additive to DME to initiate CH_4 conversion at low temperatures [110]. It was also found that CH_2O is nearly exclusively formed from DME, whereas CH_3OH is partly produced from CH_4 at temperatures around 800 K. At a temperature of 623 K, at which the maxima of CH_2O and CH_3OH mole fractions were observed in the 10 % DME mixtures, CH_4 contributes greatly to the production of both species. The reason for this behavior is that more OH radicals are produced due to the oxidation of DME, which react with CH_4 and, in connection with a low temperature, favor the production of both oxygenated species. In contrast, at higher temperatures pathways from DME to CH_2O, CH_3OH and its precursors are favored, decreasing the product contribution of CH_4. Unsaturated hydrocarbons like C_2H_4 and C_3H_6 were found to be mainly produced from C_2H_6 and C_3H_8, respectively. Considerable yields (related to the total amount of carbon in the mixture) of ~10 % (C_2H_4) and ~1 % (C_3H_6) were found with respect to these species, considering the low initial mole fractions of C_2H_6 and C_3H_8.

Model predictions of the modified reaction mechanism and another state-of-the-art reaction mechanism are in good agreement with the experimental data and have proven useful to identify suitable conditions for the formation of useful chemicals. Some disagreement between simulation and experiments corroborate the need for further improvement of both mechanisms. This need is reinforced by high-pressure shock tube measurements. The mole fractions of benzene could only be predicted well by PolyMech2.0 and in the case of pure natural gas by the Cai and Pitsch mechanism [99]. At these high-pressure conditions and higher temperatures compared to the flow reactor the influence of the natural gas components on the product composition was much less pronounced. The product composition is almost identical to experiments with methane or methane/additive (*n*-heptane, diethyl ether, dimethoxymethane) mixtures [100,101] as fuel. Ignition delay time measurements showed that the reactivity of the natural gas/air mixture is significantly increased by DME addition and that natural gas is much more reactive compared to methane also at these very fuel-rich conditions. The AramcoMech3.0 [42] predicts the measured IDTs very well whereas the PolyMech2.0 predicts slightly too long IDTs. Regarding the prediction capabilities for ignition delay times and product formation and its low number of species and reactions the PolyMech2.0 is currently the best choice for simulating and optimizing polygeneration processes with natural gas/DME as fuel at very fuel-rich conditions in IC engines.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.10.004.

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